

Longitudinal Data Analysis with R Statistical Software

Day 1. Introduction to Clustered and Longitudinal Data Analysis

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Welcome and Introductions

Overview of the Workshop

- Day 1: Introduction to Clustered and Longitudinal Data, simple analysis approaches
- Day 2: Generalized estimating equations (GEE)
- Day 3: Generalized linear mixed effects models (GLMM)
- Day 4: More on GEE and GLMM
- Day 5: Joint modeling of longitudinal and survival data

Overview of the Workshop (continued)

- Lectures
- Simulation Exercises
- Data analysis using synthetic HIV cohort data

Overview of Day 1

Clustered data

Longitudinal data

Overview

Clustered data

Longitudinal data

Correlated data

- Clustered data
 - ▶ Patients within health-care centers
- Longitudinal data
 - ▶ Patients over time
- Clustered longitudinal data
 - ▶ Patients over time within health-care centers
 - ▶ Patients over time within surgeons within health-care centers

Clustered Data

Data collected in groups, generally resulting in correlation between observations in same group

Examples

- Surveys conducted within families
- Students within classrooms
- Patients within health facilities
- Multiple observations taken on the same person

Clustered Data

Correlation between observations in same cluster should be addressed.

Hypothetical Systolic Blood Pressure (mmHG) Example:

- Blood pressure measured twice on each person

ID	treatment	BP 1	BP 2
1	0	122	123
2	0	120	119
3	0	119	119
4	0	116	118
5	0	125	123
6	1	115	115
7	1	110	112
8	1	121	120
9	1	124	123
10	1	111	111

Clustered Data

There were 10 people in the study.

Let's do a t-test comparing the first observation on each of these people:

```
##  
##  Welch Two Sample t-test  
##  
## data:  y1 by x  
## t = 1.3416, df = 6.2002, p-value = 0.2268  
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to 0  
## 95 percent confidence interval:  
##   -3.400485  11.800485  
## sample estimates:  
## mean in group 0 mean in group 1  
##           120.4           116.2
```

Clustered Data

There were 20 measurements in the study.

Let's do a t-test comparing the 20 measurements:

```
##  
##  Welch Two Sample t-test  
##  
## data:  c(y1, y2) by c(x, x)  
## t = 2.2068, df = 13.465, p-value = 0.04525  
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to  
## 95 percent confidence interval:  
##  0.1027358 8.2972642  
## sample estimates:  
## mean in group 0 mean in group 1  
##           120.4           116.2
```

Clustered Data

The second analysis is incorrect.

- Even though there were 20 observations in the study, observations on the same individual are highly correlated.
- The sample size is not 20 independent observations
- The t-test, and many other basic statistical procedures, assume independence between observations
- There are 10 independent subjects, each measured twice
- The first analysis is correct, in the sense that it only includes independent observations
- However, the first analysis ignored half of the observations, so we are losing information

Clustered Data

Another analysis:

Take the mean blood pressure for each individual and compare means:

```
##  
##  Welch Two Sample t-test  
##  
## data:  (y1 + y2)/2 by x  
## t = 1.4896, df = 5.8758, p-value = 0.1879  
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to 0  
## 95 percent confidence interval:  
##  -2.734748 11.134748  
## sample estimates:  
## mean in group 0 mean in group 1  
##           120.4           116.2
```

Clustered Data

A potential analysis strategy with clustered data:

- Obtain a summary measurement per independent observation
- Perform analysis on summary measures
- We call this a derived variables analysis
- Not a bad approach, but there are better methods
- What happens if there are 5 observations on one person and 1 observation on another person?
- We will highlight better approaches (GEE and mixed effects models later in the workshop)

Overview

Clustered data

Longitudinal data

Longitudinal studies

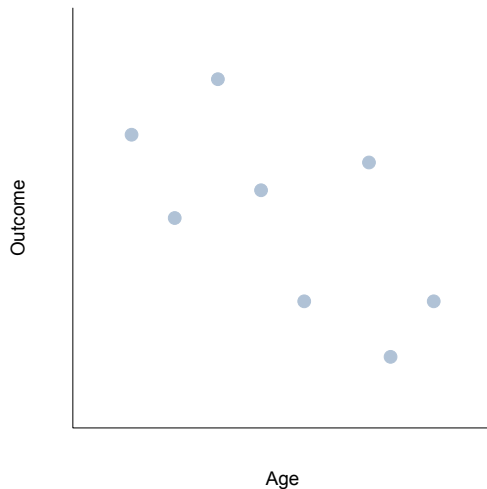
Repeatedly collect information on the same individuals over time

Benefits

- Record incident events
- Ascertain exposure prospectively
- Identify time effects: cohort, period, age
- Summarize changes over time within individuals
- Help establish causal effect of exposure on outcome
- Offer attractive efficiency gains over cross-sectional studies

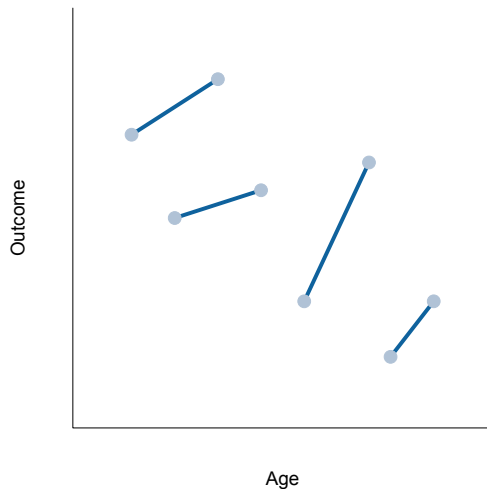
Longitudinal studies

Identify time effects: cohort, age



Longitudinal studies

Identify time effects: cohort, age



Longitudinal studies

Identify time effects: cohort, period, age

- Cohort effects
 - ▶ Differences between individuals at baseline
 - ▶ “Level”
 - ▶ Cartoon example: Younger individuals begin at a higher level
- Age effects
 - ▶ Differences within individuals over time
 - ▶ “Trend”
 - ▶ Cartoon example: Outcomes increase over time for everyone
- Period effects may also matter if measurement date varies

Longitudinal studies

Summarize changes over time within individuals

- We can partition age into two components
 - ▶ Cross-sectional (baseline) comparison

$$E[Y_{i1}] = \beta_0 + \beta_C x_{i1}$$

- ▶ Longitudinal (change) comparison

$$E[Y_{ij} - Y_{i1}] = \beta_L(x_{ij} - x_{i1})$$

for observation $j = 1, \dots, m_i$ on individual $i = 1, \dots, n$

- Putting these two models together we obtain

$$E[Y_{ij}] = \beta_0 + \beta_C x_{i1} + \beta_L(x_{ij} - x_{i1})$$

- β_L represents the expected change in the outcome per unit change in x

Longitudinal studies

Help establish causal effect of exposure on outcome

- Cross-sectional study

Egg → Chicken
Chicken → Egg

- Longitudinal study

Bacterium → Dinosaur → Chicken

- ★ There are several other challenges to generating causal inference from longitudinal data, particularly observational longitudinal data

Longitudinal studies (Old slide)

Offer attractive efficiency gains over cross-sectional studies

- Cross-sectional comparison of treatments A and B

$$\begin{aligned}\hat{\Delta} &= \bar{Y}_1^A - \bar{Y}_1^B \\ \text{Var}[\hat{\Delta}] &= 2\sigma^2 / n,\end{aligned}$$

- Longitudinal comparison of treatments A and B

$$\begin{aligned}\hat{\Delta}^* &= (\bar{Y}_1^A - \bar{Y}_0^A) - (\bar{Y}_1^B - \bar{Y}_0^B) \\ \text{Var}[\hat{\Delta}^*] &= 2\sigma^2(1 - \rho) / n\end{aligned}$$

- Longitudinal estimate may be more precise

$$\begin{aligned}0 < \rho < 1 &\Rightarrow \text{Var}[\hat{\Delta}] > \text{Var}[\hat{\Delta}^*] \\ \rho = 0.5 &\Rightarrow \text{Var}[\hat{\Delta}] = 2\text{Var}[\hat{\Delta}^*]\end{aligned}$$

- May ameliorate bias because each individual “acts as their own control”

Longitudinal studies

Potentially offer attractive efficiency gains over cross-sectional studies.

Let $\text{Var}(Y_1^A) = \text{Var}(Y_1^B) = \text{Var}(Y_0^A) = \text{Var}(Y_0^B) = \sigma^2$, $\text{Corr}(Y_0, Y_1) = \rho$, and n be the number in treatments A and B .

- Cross-sectional comparison of treatments A and B :

$$\begin{aligned}\hat{\Delta} &= \bar{Y}_1^A - \bar{Y}_1^B \\ \text{Var}[\hat{\Delta}] &= 2\sigma^2 / n\end{aligned}$$

- Longitudinal comparison of treatments A and B :

$$\begin{aligned}\hat{\Delta}^* &= (\bar{Y}_1^A - \bar{Y}_0^A) - (\bar{Y}_1^B - \bar{Y}_0^B) \\ \text{Var}[\hat{\Delta}^*] &= \text{Var}[(\bar{Y}_1^A - \bar{Y}_0^A) - (\bar{Y}_1^B - \bar{Y}_0^B)] \\ &= \text{Var}(\bar{Y}_1^A) + \text{Var}(\bar{Y}_0^A) + \text{Var}(\bar{Y}_1^B) + \text{Var}(\bar{Y}_0^B) - 2\text{Cov}(\bar{Y}_1^A, \bar{Y}_0^A) - 2\text{Cov}(\bar{Y}_1^B, \bar{Y}_0^B) \\ &= 4\sigma^2 - 4\sigma^2\rho \\ &= 4\sigma^2(1 - \rho) / n\end{aligned}$$

Longitudinal studies

Potentially offer attractive efficiency gains over cross-sectional studies

- Cross-sectional comparison of treatments A and B :

$$\begin{aligned}\hat{\Delta} &= \bar{Y}_1^A - \bar{Y}_1^B \\ \text{Var}[\hat{\Delta}] &= 2\sigma^2 / n\end{aligned}$$

- Longitudinal comparison of treatments A and B :

$$\begin{aligned}\hat{\Delta}^* &= (\bar{Y}_1^A - \bar{Y}_0^A) - (\bar{Y}_1^B - \bar{Y}_0^B) \\ \text{Var}[\hat{\Delta}^*] &= 4\sigma^2(1 - \rho) / n\end{aligned}$$

- Longitudinal estimate may be more precise

$$\begin{aligned}0.5 < \rho < 1 &\Rightarrow \text{Var}[\hat{\Delta}] > \text{Var}[\hat{\Delta}^*] \\ \rho = 0.5 &\Rightarrow \text{Var}[\hat{\Delta}] = \text{Var}[\hat{\Delta}^*]\end{aligned}$$

- May ameliorate bias because each individual “acts as their own control”

Longitudinal studies

Repeatedly collect information on the same individuals over time

Challenges

- Account for incomplete participant follow-up
- Determine causality when covariates vary over time
- Choose exposure lag when covariates vary over time
- Require specialized methods that account for longitudinal correlation

Longitudinal studies

Require specialized methods that account for longitudinal correlation

- Individuals are assumed to be independent
- Longitudinal dependence is typically a secondary/nuisance feature
- Ignoring dependence may lead to incorrect inference
 - ▶ Longitudinal correlation usually positive
 - ▶ Estimated standard errors may be too small
 - ▶ Confidence intervals are too narrow; too often exclude true value

Simulation Exercise 1

Generate $X_i = 0$ for $i = 1, \dots, n/2$, and $X_i = 1$ for $i = n/2 + 1, \dots, n$, denoting control or treatment, respectively; thus, the first half of observations get control and the second half of observations get treatment.

Generate pre- and post-baseline measures of an outcome, Y_{0i} and Y_{1i} , respectively, where $(Y_0, Y_1) \sim N(\mu, \Sigma) + (0, \beta X)$, with $\mu = (5, 5)$, $\Sigma = \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix}$, $n = 100$, $\rho = 0.5$, and $\beta = 0.5$.

Hint: Use the `mvrnorm` function in the MASS library.

1. Compare estimates for the following analysis strategies:

- A. t-test comparing Y_1 for $X = 0$ vs. $X = 1$
- B. t-test comparing $Y_1 - Y_0$ for $X = 0$ vs. $X = 1$
- C. linear model regressing Y_1 on X
- D. linear model regressing Y_1 on X and Y_0 (ANCOVA)
- E. linear model regressing $Y_1 - Y_0$ on X
- F. linear model regressing $Y_1 - Y_0$ on X and Y_0 (ANCOVA)

2. Repeat 1000 times to estimate the power of each analysis strategy. Vary $\rho \in \{0, 0.25, 0.50, 0.75, 0.90\}$. What do you learn?

Simulation Exercise 1

Code for generating data:

```
library(MASS)

rm(list=ls())

n<-100
rho<-0.5
beta<-0.5
y<-mvrnorm(n, mu=c(5,5), Sigma=matrix(c(1,rho,rho,1),ncol=2))
y0<-y[,1]
x<-c(rep(0,n/2),rep(1,n/2))
y1<-y[,2]+beta*x
```

Simulation Exercise 1

Code for fitting models:

```
diff<-y1-y0  
  
t.test(y1~x)  
t.test(diff~x)  
summary(lm(y1~x))  
summary(lm(y1~x+y0))  
summary(lm(diff~x))  
summary(lm(diff~x+y0))
```

Simulation Exercise 1

Code for running a simulation loop:

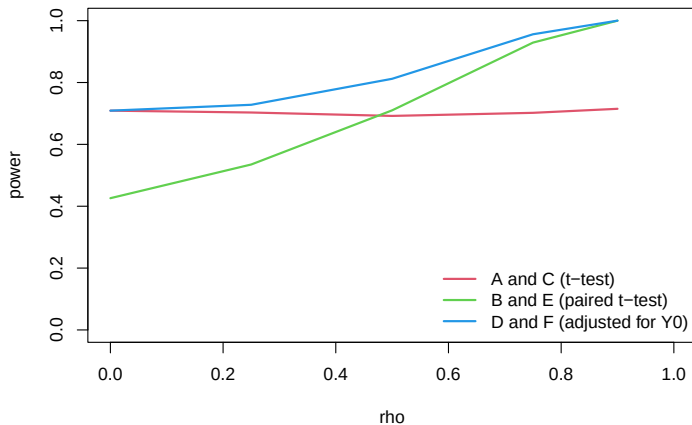
```
nsims<-1000
rho<-0.5

pvalA<-pvalB<-pvalC<-pvalD<-pvalE<-pvalF<-NULL
estC<-estD<-estE<-estF<-NULL
for (i in 1:nsims) {

  ## Generate the data
  ## Fit the models
  ## Save results:
  pvalA[i]<-t.test(y1~x)$p.value      #example
  pvalC[i]<-summary(modC)$coeff[2,4] #example
}

mean(pvalA<.05)
mean(pvalC<.05)
```

Simulation Exercise 1

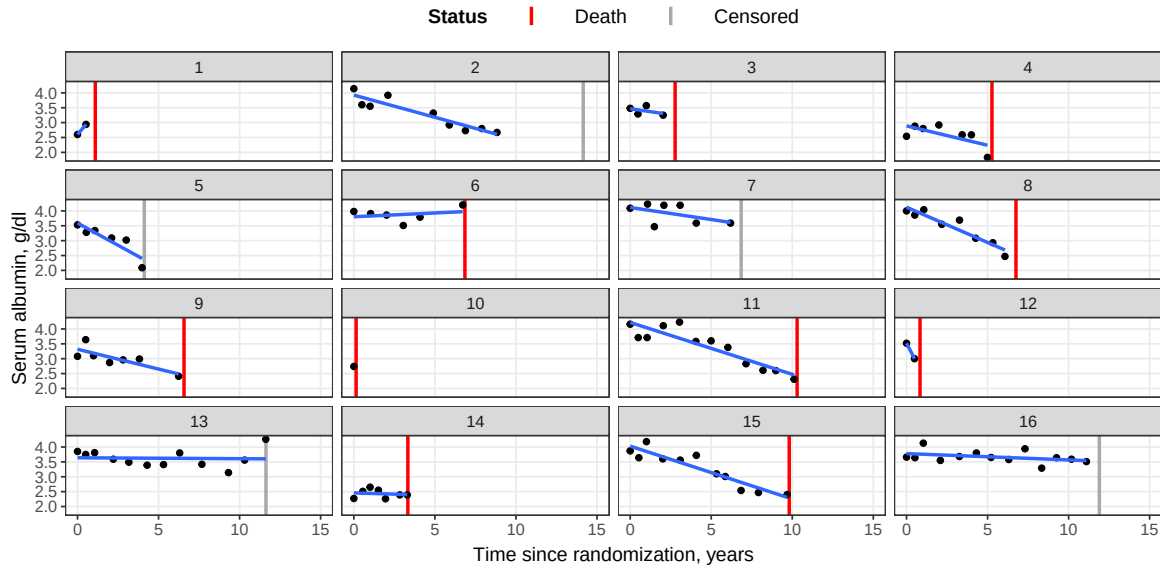


PBC data

Mayo Clinic trial in primary biliary cirrhosis (Murtaugh et al., 1994)

- Primary biliary cirrhosis (or, cholangitis)
 - ▶ Chronic and fatal but rare autoimmune disease
 - ▶ Inflammatory destruction of small bile ducts within the liver
- Randomized trial of immunosuppressive therapy
 - ▶ Patients referred to Mayo Clinic, 1974–1984
 - ▶ 158 patients randomized to treatment with D-penicillamine; 154 randomized to placebo
- Quantify covariate associations with biomarker levels over time (longitudinal)
 - ▶ Serum albumin: Crucial for fluid balance, with low values indicative of reduced liver and/or kidney function

PBC data

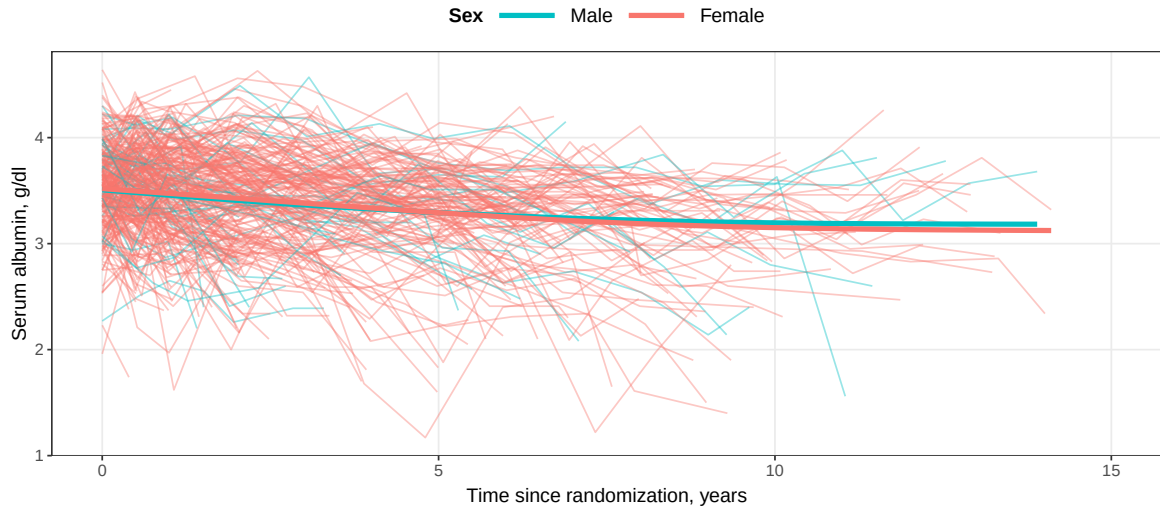


PBC data

Code that was used for the previous figure:

```
ggplot(subset(pbc2, is.element(id, 1:16)), aes(year, albumin)) +  
  facet_wrap( ~ id) +  
  geom_vline(aes(xintercept=years, color=factor(status!='dead')), lwd=1) +  
  geom_point() +  
  geom_smooth(se=FALSE, method='lm') +  
  scale_x_continuous(name='Time since randomization, years', limits=c(0,15)) +  
  scale_y_continuous(name='Serum albumin, g/dl') +  
  scale_colour_manual(name='Status', breaks=c(FALSE, TRUE), values=c('red', 'darkgray'),  
                      labels=c('Death', 'Censored')) +  
  mytheme()
```

PBC data



PBC data

Code that was used for the previous figure:

```
data(pbc2)
pbc2 <- within(pbc2, {
  levels(sex) <- c("Male", "Female")} )
ggplot(subset(pbc2, albumin<6), aes(year, albumin, group=id)) +
  geom_line(aes(color=sex), alpha=0.4) +
  geom_smooth(aes(year, albumin, group=sex, color=sex), se=FALSE, linewidth=1.5) +
  scale_x_continuous(name='Time since randomization, years', limits=c(0,15)) +
  scale_y_continuous(name='Serum albumin, g/dl') +
  scale_color_manual(name='Sex', values=c('#00BFC4', '#F8766D')) +
  mytheme()
```

Perhaps use the observations at time 0 and time 1 to illustrate similarities between paired t-test, adjusted analysis, time 1 analysis (t-test), etc.

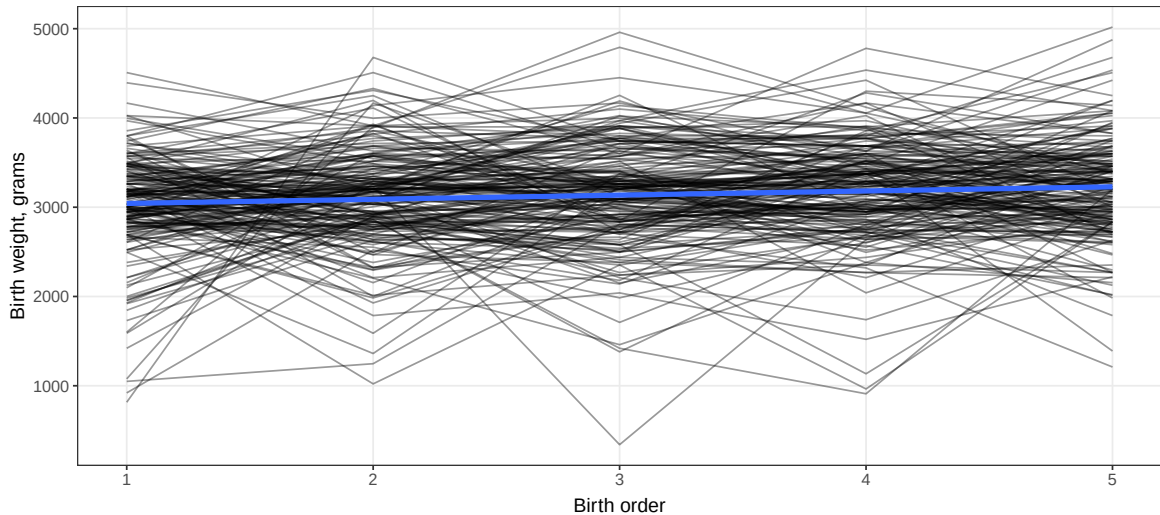
Or perhaps do this with a simulation.

Birth weight data

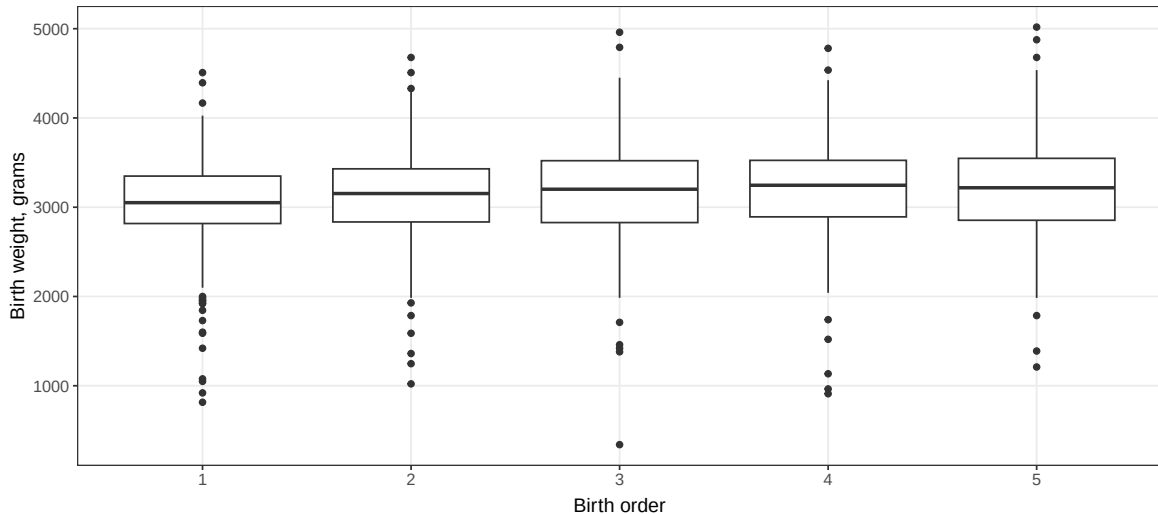
Georgia infant birth weight

- Birth weight (grams) measured for each of 5 infants born to 200 mothers
- Interested in the association between birth order (1,2,3,4,5) and birth weight
 - ▶ Estimate the average time course among all mothers
 - ▶ Estimate the time course for individual mothers
 - ▶ Quantify the degree of heterogeneity across mothers
 - ▶ Determine whether the mother's initial age (age at first birth) impacts the time course

Birth weight data



Birth weight data



Potential Analysis Approach

- **Derived variables:** Collapse the longitudinal series for each individual into a summary statistic and use analysis methods for independent data
 - ▶ Difference: birth weight for last infant – birth weight for first infant
 - ▶ Slope: linear regression model for each individual

Derived variables

Collapse the longitudinal series for each individual into a summary statistic and use analysis methods for independent data

- Approach is typically straightforward to implement and understand (+)
 - ▶ Calculate a change over time for each individual (e.g., difference, slope)
 - ▶ Summarize the change across individuals
 - ▶ Correlate the change with individual-level covariates
- But has several limitations (—)
 - ▶ Can understate variability
 - ▶ Can be impacted by missing data
 - ▶ Unclear how to address covariates that change over time

Birth weight data: Derived variables

Difference in weight between first and last baby:

```
bwt |>
  group_by(momid) |>
  mutate(bwdiff=bweight[birthord==5]-bweight[birthord==1]) |>
  filter(row_number()==1) |>
  ungroup() -> temp

head(temp, 5)
```

```
## # A tibble: 5 x 5
##   momid birthord initage bweight bwdiff
##   <int>    <int>   <int>   <int>   <int>
## 1     39         1     15    3720   -1240
## 2     62         1     17    2381    -170
## 3     79         1     17    2940  -1551
## 4     80         1     18    3175     652
## 5     84         1     14    2892     510
```

Birth weight data: Derived variables

Do baby's weights increase with birth order? (Comparing if difference in first and last babies' weights differ from zero.)

```
with(temp, summary(bwdiff))
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## -1551.0  -190.0   164.0   191.6   482.0  2700.0
```

```
with(temp, t.test(bwdiff))
```

```
##
##  One Sample t-test
##
## data:  bwdiff
## t = 4.211, df = 199, p-value = 3.849e-05
## alternative hypothesis: true mean is not equal to 0
## 95 percent confidence interval:
##  101.898 281.382
## sample estimates:
## mean of x
```

Birth weight data: Derived variables

Is the change in babies' weights based on birth order correlated with the mother's age at first pregnancy?

```
with(temp, cor.test(bwdiff, initage))
```

```
##  
## Pearson's product-moment correlation  
##  
## data: bwdiff and initage  
## t = 0.62787, df = 198, p-value = 0.5308  
## alternative hypothesis: true correlation is not equal to 0  
## 95 percent confidence interval:  
## -0.09475081 0.18219044  
## sample estimates:  
## cor  
## 0.04457619
```

Birth weight data: Derived variables

Summarizing using slope between birth weight and birth order within mothers:

```
bwt |>
  group_by(momid) |>
  do(tidy(lm(bweight ~ I(birthord-1), .))) -> regs

temp$beta0 <- regs$estimate[regs$term=='(Intercept)']
temp$beta1 <- regs$estimate[regs$term=='I(birthord - 1)']

head(temp, 5)
```

```
## # A tibble: 5 x 7
##   momid birthord initage bweight bwdiff beta0  beta1
##   <int>    <int>   <int>   <int>   <int> <dbl>  <dbl>
## 1     39        1     15    3720  -1240 3822   -242
## 2     62        1     17    2381   -170 2597.  -90.7
## 3     79        1     17    2940  -1551 3126. -222.
## 4     80        1     18    3175    652 3232.  201.
## 5     84        1     14    2892    510 3187   76.4
```

Birth weight data: Derived variables

Does slope differ from zero?

```
with(temp, summary(beta1*5))
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## -1361.5  -152.1   213.0   233.0   528.0  2420.0
```

```
with(temp, t.test(beta1))
```

```
##
##  One Sample t-test
##
## data:  beta1
## t = 4.6509, df = 199, p-value = 6.017e-06
## alternative hypothesis: true mean is not equal to 0
## 95 percent confidence interval:
##  26.84635 66.36965
## sample estimates:
## mean of x
##    46.608
```

Birth weight data: Derived variables

Is slope correlated with maternal age at first pregnancy?

```
with(temp, cor.test(beta1, initage))

##
## Pearson's product-moment correlation
##
## data: beta1 and initage
## t = -0.3155, df = 198, p-value = 0.7527
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## -0.1606576 0.1166877
## sample estimates:
## cor
## -0.02241619
```


Analysis approaches

- **Derived variables:** Collapse the longitudinal series for each individual into a summary statistic and use analysis methods for independent data (+/−)
- **Repeated measures:** Include all data in a regression model for the mean response and account for longitudinal and/or cluster correlation
 - ▶ **Generalized estimating equations (GEE):** A marginal model for the mean response and a model for longitudinal or cluster correlation

$$g(E[Y_{ij} | x_{ij}]) = x_{ij}\beta \quad \text{and} \quad \text{Corr}[Y_{ij}, Y_{ij'}] = \rho(\alpha), j \neq j'$$

- ▶ **Mixed-effects models:** A conditional model for the mean response given subject-specific random effects, which induce a (possibly hierarchical) correlation structure

$$g(E[Y_{ij} | x_{ij}, b_i]) = x_{ij}\beta + z_{ij}b_i \quad \text{with} \quad b_i \sim \text{Normal}(0, D)$$

NB: Potential difference in interpretation for β between GEE and mixed-effects models

Format of Longitudinal Data

- Wide Format

ID	sex	t_1	t_2	t_3	t_4	t_5
1	M	3.8	3.1	2.0	NA	NA
2	M	4.1	3.5	3.8	2.4	2.8
3	F	2.7	2.4	2.9	3.5	NA

Format of Longitudinal Data

- Long Format

ID	sex	time	outcome
1	M	1	3.8
1	M	2	3.1
1	M	3	2.0
2	M	1	4.1
2	M	2	3.5
2	M	3	3.8
2	M	4	2.4
2	M	5	2.8
3	F	1	2.7
3	F	2	2.4
3	F	3	2.9
3	F	4	3.5

Summary of Day 1

- Clustered data
- Longitudinal data
- Importance of acknowledging correlation in analyses
- Derived variables analysis strategies